ASSIGNMENT 2, LLM Prompting

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Abstract

This report presents a comparative analysis of two state-of-the-art open-source Large Language Models, **Meta Llama-3.1-8B-Instruct** and **AllenAI Olmo-7B-Instruct**, applied to the EDOS Task B (fine-grained sexism detection). We evaluated the models in a resource-constrained environment utilizing 4-bit NormalFloat (NF4) quantization. By experimenting with Zero-shot and Few-shot prompting strategies, we assess the models' In-Context Learning (ICL) capabilities. Our results demonstrate that Llama-3.1 significantly outperforms Olmo, achieving a Macro-F1 score of **0.4806** in the Few-shot setting. While Llama effectively leverages demonstrations to refine its decision boundaries, Olmo exhibits a strong conservative bias, often collapsing predictions to the majority class despite the provision of in-context examples.

1 Introduction

Online sexism detection is a critical area in Natural Language Processing (NLP), necessitating models capable of discerning subtle linguistic nuances. This study addresses **Task B** of the Explainable Detection of Online Sexism (EDOS) challenge(Kirk et al., 2023), a multi-class classification problem involving five distinct labels: *not-sexist, threats, derogation, animosity,* and *prejudiced discussions*. Distinguishing between categories such as *animosity* and *derogation* is particularly challenging due to their semantic overlap.

Recent advancements in Large Language Models (LLMs) suggest that generative approaches can compete with traditional discriminative classifiers. However, deploying 7B+ parameter models presents significant hardware challenges. We aim to benchmark the widely adopted **Llama-3.1-8B** against the fully open-source **Olmo-7B**, investigating whether quantization and few-shot prompting can yield competitive performance without fine-tuning.

2 System Description

2.1 Model Architectures

We utilized two state-of-the-art instruction-tuned models available on the Hugging Face Hub, ensuring a comparison between contemporary open-weights architectures:

- **Meta Llama-3.1-8B-Instruct**: Based on a decoder-only transformer architecture, it utilizes Grouped-Query Attention (GQA) and is optimized via Direct Preference Optimization (DPO). It represents the industrial state-of-the-art for 8B parameter models (AI@Meta, 2024).
- **AllenAI Olmo-3-7B-Instruct**: The latest iteration of the fully open Olmo family. Unlike Llama, Olmo releases full training data and checkpoints to promote scientific transparency. We assess whether this newest version bridges the performance gap with Llama 3.1 in complex classification tasks (Groeneveld et al., 2024).

2.2 Quantization and Hardware

Experiments were conducted on a single T4 GPU (Google Colab). To fit the models within the ≈15GB VRAM limit, we employed the `bitsandbytes` library for **4-bit quantization**(Dettmers and Pagnoni, 2023). Specifically, we used the **NF4 (NormalFloat 4)** data type, which is information-theoretically optimal for normally distributed weights, allowing us to load the ∼16GB fp16 weights in under 6GB of VRAM with negligible degradation in reasoning performance.

2.3 Prompting Strategies

We designed a rigid template to ensure parsing reliability. The system prompt defines the 5 classes and instructs the model to output a single label.

- **Zero-Shot:** The model receives the instruction and the input text. This tests the model's intrinsic knowledge acquired during pre-training.
- **Few-Shot (In-Context Learning):** We augmented the prompt with $k = 4$ demonstrations per class (20 shots total). Exam-

ples were retrieved from a static support set (`demonstrations.csv`). This setting tests the model's ability to infer the task distribution from the context window without weight updates.

3 Experimental Setup

3.1 Generation Parameters

Reproducibility is paramount. We disabled sampling by setting the temperature $T = 0.0$ and using **Greedy Decoding**. This ensures that the model outputs the most probable token sequence deterministically.

3.2 Parsing and Metrics

The raw string output is parsed against the set of valid labels. We implemented a "Fail Ratio" metric to track hallucinations (outputs not matching any class). Given the class imbalance inherent in hate speech datasets, we report the **Macro-F1 Score** as the primary performance metric, as Accuracy would be misleadingly high due to the dominance of the *not-sexist* class.

4 Results

Table 1 summarizes the experimental results. Both models achieved a 0% Fail Ratio, indicating that the instruction-tuned nature of these models is sufficient to follow output format constraints.

Model	Strategy	Macro-F1
Olmo-7B-Instruct	Zero-Shot	0.3520
Olmo-7B-Instruct	Few-Shot	0.3626
Llama-3.1-8B-Instruct	Zero-Shot	0.4550
Llama-3.1-8B-Instruct	**Few-Shot**	**0.4806**

Table 1: Comparison of Macro-F1 scores. Llama-3.1 outperforms Olmo by a significant margin (+11.8 points in Few-Shot).

Llama-3.1 demonstrates superior performance across the board. Notably, the transition from Zero-shot to Few-shot yields a performance gain of approximately **+2.5%** for Llama, whereas Olmo's improvement is marginal ($\approx 1\%$).

5 Discussion

5.1 In-Context Learning Effectiveness

The results highlight a divergence in In-Context Learning (ICL) abilities. Llama-3.1 effectively utilizes the 20 provided examples to better distinguish between semantically similar classes (e.g., *prejudiced* vs. *derogation*). This suggests that Llama's attention heads are more capable of retrieving relevant information from the extended context. Conversely, Olmo-7B appears to struggle with integrating the demonstrations, showing a "saturation" effect where additional context does not translate to better classification boundaries.

5.2 The "Conservative Bias" in Olmo

A qualitative analysis of the error logs reveals a distinct behavior in Olmo-7B: a **conservative bias**. In the presence of uncertainty, particularly in the Few-shot setting, Olmo tends to default to the *not-sexist* label (False Negatives). For instance, explicit threats such as *"She deserves to be beaten"* were occasionally classified as *not-sexist* by Olmo. We hypothesize this may be a side-effect of its safety alignment process (RLHF), which might suppress the generation of toxic labels even when the task requires classifying toxic content. Llama-3.1, conversely, maintained a higher recall on minority classes, proving more robust for safety-critical tasks like hate speech detection.

5.3 Limitations

While 4-bit quantization enabled this study, it introduces a small quantization error. Furthermore, fixed demonstrations might not be optimal for every input; future work could explore *dynamic few-shot selection* (using similarity search) to retrieve the most relevant examples for each specific input tweet.

6 Conclusion

Our comparative analysis confirms that **Meta Llama-3.1-8B** is currently the superior open-weight choice for the EDOS task, significantly outperforming Olmo-7B. Llama's architecture and training data provide a better foundation for few-shot learning, allowing it to navigate the complexities of sexism detection with higher F1 scores. Olmo, while a valuable scientific artifact, requires further refinement or fine-tuning to mitigate its conservative bias in zero-shot scenarios.

References

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